Listing 3.8 method Treap::min

```
function min(treap)
                            #1
  throw-if treap.root == null
                                                  #2
  node ← treap.root
                                               #3
  while node.left != null
do                                                    #4
    node ← node.left
  return
node.key
#5
```

#1 Method `min` takes a treap instance and returns its top-priority element, unless the treap is empty.
#2 If the treap was empty, we need to throw an error (it would have no min, obviously).
#3 Initialize the temporary variable node with the tree's root (not null, because of the check at line 2)
#4 Until we reach the left-most node, keep traversing the left branch.
#5 Return the node's key.

3.3.8 Performance Recap

This concludes our discussion on the implementation of treaps; in the next sections, we'll discuss applications of treaps and analyze them in more details.

For now, let's recap the running time of the treap's methods, shown in table 3.2; notice that

- All operations only depend on the height of the tree, rather than on the number of elements. Of course, in the worst-